

RESEARCH PROPOSAL

Developing AI as a Transformative Global Catalyst: The $C = G \times I^2$ Framework
for Human-Predefined Outcomes, Autonomous Mechanism Synthesis, and
Long-Term Cascading Effects Aligned with UN SDG17

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Project Period: 1 July 2026 – 30 June 2028 (24 months)

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1. INTRODUCTION

The rapid advancement of Artificial Intelligence presents both unprecedented opportunities and significant risks of misalignment. This research proposes the $C = G \times I^2$ Framework as a mathematically rigorous, architecturally implementable, and ethically grounded model to develop AI as a beneficial, transformative global catalyst, strongly aligned with United Nations Sustainable Development Goal 17 (Partnerships for the Goals).

2. TECHNICAL ABSTRACT & CORE FRAMEWORK

Abstract

Title: Developing AI as a Transformative Global Catalyst Framework Aligned with UN SDGs

Abstract:

Artificial Intelligence (AI) has emerged as a critical driver for global development, yet a unified approach to maximizing its impact across the United Nations Sustainable Development Goals (UN SDGs) remains largely unaddressed. This paper introduces a novel strategic framework designed to operationalize AI as a transformative catalyst for global sustainability. At the core of this framework is the conceptual equation $C = G \times I^2$, where Catalytic Impact (C) is determined by the synergy between Global Governance (G) and the exponential power of Innovation and Integration (I^2). This study investigates how structured governance can safely steer computing infrastructure and deep learning models to address pressing socio-economic and environmental challenges. By establishing this mathematical and theoretical architecture, the research provides a scalable blueprint for policymakers, tech developers, and global institutions to align autonomous systems with human-centric sustainable targets. Ultimately, this framework bridges the gap between theoretical AI capabilities and practical, ethically-governed global transformations.

Keywords: Artificial Intelligence, United Nations SDGs, Global Governance, Strategic Framework, Sustainable Innovation, Technological Catalyst.

Core Equation:

$$C = G \times I^2$$

Component Definitions:

- C = Catalyst Impact: Measurable, scalable, positive systemic change.
- G = Human-Predefined Goals: Ethically grounded objectives anchored in SDG17 and universal human values.
- I = Intelligence: Advanced AI capabilities including reasoning, creativity, planning, and causal inference.
- I^2 = Intelligence × Integration: Exponential amplification when intelligence is deeply embedded across data ecosystems, organizations, and societies.

Core Dimensions:

1. Human-Prescribed Outcomes
2. Autonomous Mechanism Synthesis
3. Long-Term Cascading and Butterfly Effects

3. RESEARCH OBJECTIVES

Primary Objective

To develop, validate, and apply the $C = G \times I^2$ Framework as a practical and responsible model for AI development aligned with SDG17.

Specific Objectives

- Establish the mathematical and theoretical foundation of the framework.
- Develop a functional prototype of the Autonomous Mechanism Synthesis Engine (AMSE) and Governance Layer.
- Build a lightweight Cascading Effect Simulator.
- Validate the framework through simulation and one focused local pilot project in Thailand.

4. DETAILED METHODOLOGY AND WORK PLAN

Phase I: Conceptual & Mathematical Foundation (July – November 2026)

Month 1–2: Project setup, comprehensive literature review, and formalization of $C = G \times I^2$

Month 3–4: Develop Ontology, Causal Diagrams, and Concept Paper

Month 5: Complete Appendix A and submit Phase I report

Phase II: System Architecture Development (December 2026 – April 2027)

Month 6–8: Design hybrid architecture and develop core components (AMSE + Governance Layer)

Month 9–10: System integration, testing, and optimization

Month 11: Complete architecture documentation and Phase II report

Phase III: Simulation & Validation (May – October 2027)

Month 12–14: Develop lightweight Digital Twin / Cascading Effect Simulator

Month 15–17: Test Domino and Butterfly Effects across selected scenarios

Month 18: Perform validation, analysis, and submit Phase III report

Phase IV: Pilot Implementation & Dissemination (November 2027 – June 2028)

Month 19–21: Develop partnerships and implement one small-scale pilot in Thailand

Month 22–23: Data collection, impact evaluation, and case study writing

Month 24: Final thesis writing, Policy Brief preparation, and defense readiness

5. RECOMMENDED TOOLS AND TECHNOLOGY STACK

Project Management & Documentation

Notion or Obsidian (primary workspace)

GitHub Private Repository (version control)

Core Development

Python 3.11+

LangChain / LlamaIndex

CrewAI or AutoGen (agent framework)

PyTorch and Hugging Face Transformers

NetworkX (graph modeling)

SymPy (symbolic mathematics)

Simulation & Analysis

Mesa or SimPy (agent-based simulation)

DoWhy / CausalNex (causal inference)

Matplotlib and Plotly (data visualization)

Writing & References

Overleaf (LaTeX)

Zotero (reference management)

Grammarly

Collaboration

Google Drive / Microsoft OneDrive

Zoom / Google Meet

6. GENERAL ANALYSIS AND POINT OF VIEW DEFENDED

The $C = G \times I^2$ Framework offers a balanced third path between overly restrictive control and unrestricted development. It maintains strong human direction at the goal level while allowing high autonomy in mechanism synthesis and systemic integration.

7. LITERATURE REVIEW

This research builds upon foundational works such as Bostrom (2014), Russell (2019), Ji et al. (2024), complex systems research from the Santa Fe Institute (Krakauer et al., 2025), and UN/ITU AI for Good reports. A critical gap exists in frameworks that simultaneously integrate robust goal-locking, autonomous mechanism synthesis, and long-term cascading effect simulation. This project directly addresses that gap.

8. APPLICATION OF THE FRAMEWORK

The framework will be applied to one priority domain in Thailand, such as disaster early warning systems or climate-resilient agriculture, while offering a scalable model for broader regional and global SDG17 partnerships.

9. PERSONAL AND PROFESSIONAL ENRICHMENT

Participation in the Atlantic International University Post-Doctoral Program will significantly enhance the researcher's interdisciplinary expertise in AI technologies, complex systems science, ethics, and global sustainable development.

10. PROJECT CHARTER

Project Title: Developing AI as a Transformative Global Catalyst: The $C = G \times I^2$ Framework

Principal Investigator: Somchet Isarangkul Na Ayudhya

Duration: 1 July 2026 – 30 June 2028

Key Deliverables

- Theoretical and mathematical foundation
- Functional core system prototype
- Lightweight simulator with validation
- One local pilot case study and final thesis

11. QUARTERLY GOALS

- Q1–Q2 (2026): Theoretical foundation and Phase I completion
- Q3–Q4 (2026–2027): System architecture development and Phase II completion
- Q5–Q6 (2027): Simulator development, validation, and Phase III completion
- Q7–Q8 (2027–2028): Pilot implementation, evaluation, thesis writing, and dissemination

12. RISK REGISTER

- Insufficient computational resources — High / High — Mitigation: Cost-effective cloud and optimization
- Delayed Advisor feedback — Medium / Medium — Mitigation: Early draft submissions
- Difficulty securing pilot partners — Medium / High — Mitigation: Focus on local Thai organizations
- Data privacy and ethical issues — Medium / High — Mitigation: Synthetic data usage
- Model misalignment — High / High — Mitigation: Strong Governance Layer
- Budget constraints — High / High — Mitigation: Modular and scalable design

13. BUDGET CONTINGENCY STATEMENT

This research project is designed with a modular, flexible, and scalable structure. Phase I will commence on 1 July 2026 using minimal resources.

In the event of funding limitations, priority will be given to the theoretical foundation, the core prototype, simulation validation, and one focused pilot in Thailand. Alternative funding sources will be actively pursued throughout the project.

14. PROGRESS REPORT TEMPLATES

Monthly Progress Report

- Executive Summary
- Key Achievements
- Progress against Quarterly Goals
- Challenges and Risk Update
- Next Month Priorities

Quarterly Progress Report

- Quarter Summary
- Goals Status
- Major Deliverables
- Challenges and Mitigation
- Next Quarter Focus

15. CONCLUSIONS

$C = G \times I^2$ provides a scalable, responsible, and viable method for developing AI. The CIG Project has an established reputation for scientific integrity and is realistic in the provision of resources; therefore, it supports SDG17 through the meaningful development of Beneficial AI.

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Krakauer, D. et al. (2025). *Emergence in Artificial Systems*. Santa Fe Institute.

United Nations & ITU. (2023–2025). *AI for Good Reports*.

APPENDICES

Appendix A: Detailed Mathematical Formalization of $C = G \times I^2$

Appendix B: Figure 1 – Comparative Impact Curves

Appendix C: Thailand Pilot Case Study Outline

Declaration: This proposal represents original scholarly work by the author.

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